Final Report: Quantum Drug Discovery for Malaria & Tuberculosis Using VQE + QML

1. Abstract

This project uses Quantum Variational Eigensolver (VQE) + Quantum Machine Learning (QSVM/QNN) to accelerate drug discovery for Malaria & TB in Africa.

2. Problem Statement

Malaria & TB demand faster molecular simulation and drug identification.

Classical computation fails due to exponential scaling.

3. Quantum Solution

- VQE for molecular energy minimization.
- QSVM/QNN for drug-likeness classification.
- QGAN for generating new molecules.

4. Methodology

Workflow:

Data → Preprocessing → Hamiltonian → VQE → QSVM/QNN → QGAN → Scoring

5. Results (Prototype)

- Molecular energy estimation using VQE(UCCSD).
- QSVM prediction of active vs inactive molecules.
- QGAN generated candidate molecules.

6. Future Work

Clinical trials, HPC integration, large molecule support.

7. Impact on Africa

Faster drug discovery reduces cost and saves millions of lives.

8. Conclusion

The hybrid QML + VQE pipeline is viable for next-gen African biomedical R&D.;